# Projects and Experiences Portfolio

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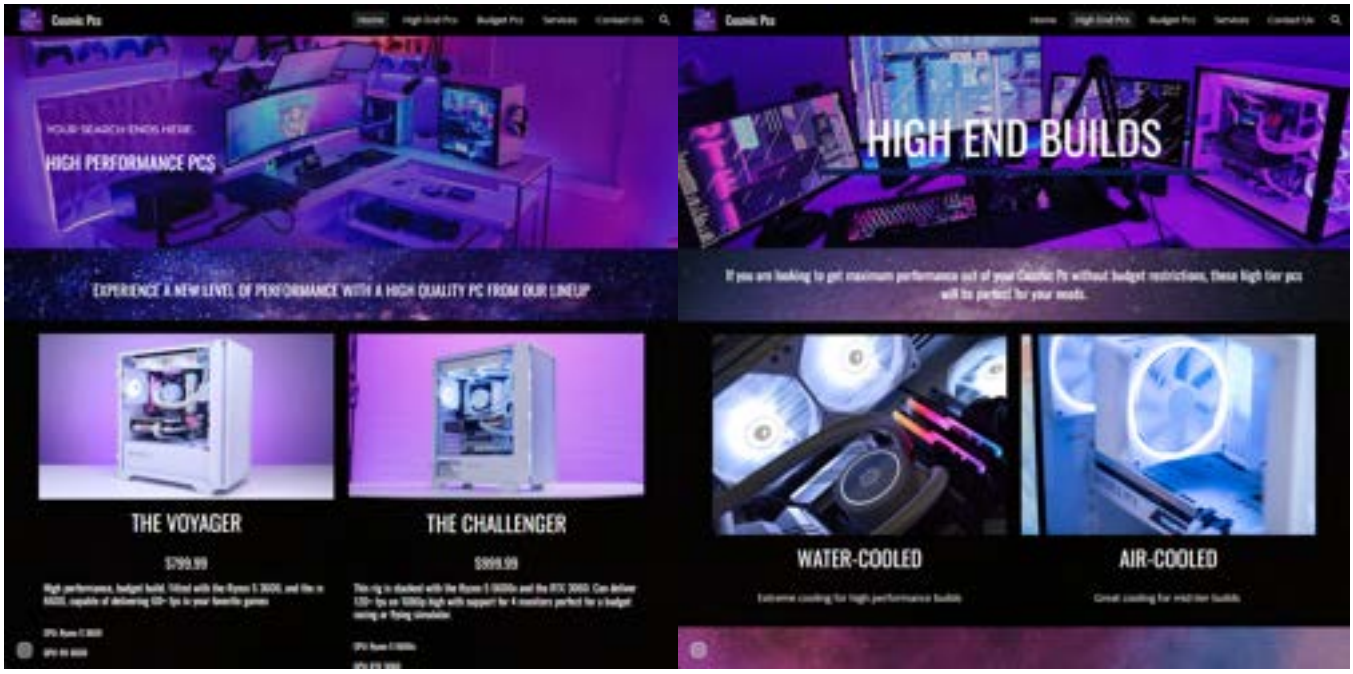
# Cosmic Pcs

Website - cosmicpcs.us




- Created an online storefront to showcase bestselling builds/configurations, created interactive buttons and popups to further allow clients to explore and understand different components within the builds.

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## Custom Parts Lists

- Worked with the team to manage and push orders, created custom parts-lists for each client, ensuring that it matches both budget and performance requirements.
- Sent the parts-lists to clients, showcased all of the selected components as well as the pros and cons of each selection. Modified the orders to the customers preferences, such as color choice, performance, and compatibility of the parts.
- Organized all orders utilizing Excel and Google Calendar, allowing the team to be on the same page when it came to processing orders, ensuring that any team member could pick up an unfinished order and know exactly where it was left off.

## The Build Station/Test Bench

- Created a fully functional build station, suitable for PC building, diagnosis, and media/content creation. Featured an operational test bench, allows for diagnosis of individual parts such as SSDs, HDDs, and GPUs. Had additional replacement parts such as various screws, OS installation drives, and an assortment of tools.

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## Custom Builds

- A handful of the most common configurations ordered by customers, some opting for watercooling over traditional air cooling.
- Some clients also bring in their existing systems for upgrades or diagnosis on certain issues. These timeslots require extra time for troubleshooting as they can range in terms of complexity, from something as simple as adding storage volumes, to diagnosing a faulty component within a system. Oftentimes an upgraded part is pushed to the client upon completion of troubleshooting, and the system is fully operational the same day to boost client satisfaction.

Social Media Outreach (Instagram)

- Created an Instagram page to interact with clients and potential customers, showcased builds as well as created build videos on a few builds to illustrate the process behind building a computer with CosmicPCs.
- Managed the page as well as increased media outreach by responding to interested clients who had questions and concerns when it came to building with us.
- Utilized a GoPro and the Adobe client to record and edit content. These videos are viewable on the instagram page: Cosmic_PCs

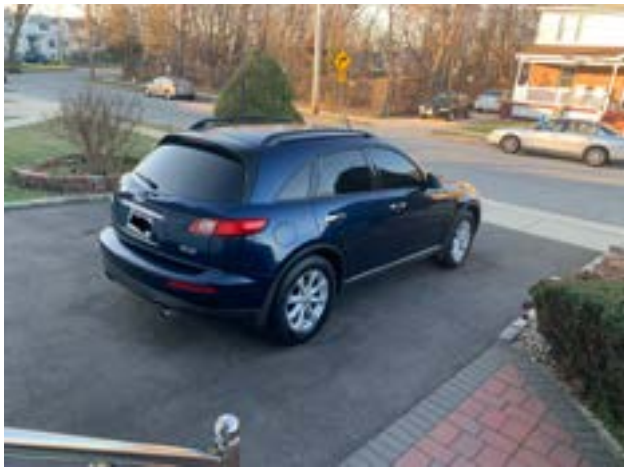
# Automotive Experience

## Buying and Selling

- Started my freshman year of highschool, saved up for my first car by working a summer job, bought my first car: 2008 Chevy Impala Police Package, did basic maintenance, sparkplugs, brakes, injectors, complete fluid flush, and a full head gasket and valve cover gasket change. Bought the car for $2000 and flipped it on a local selling platform (Facebook Marketplace) for a profit.

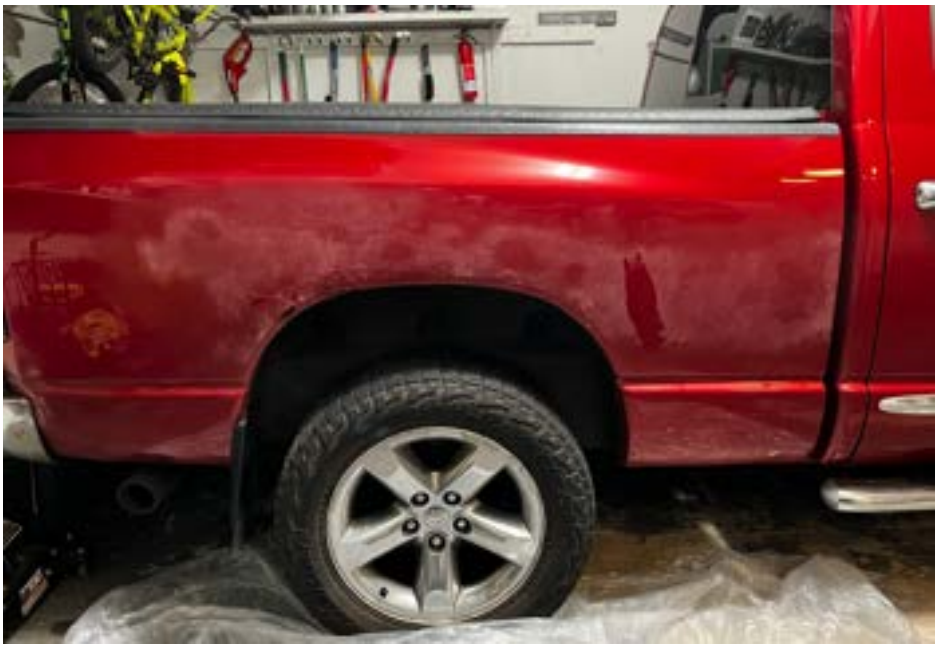
- After flipping this car, I bought my next car off of my uncle, a 2013 Toyota Corolla, which underwent the same maintenance process as the Impala, gaining automotive experience along the way.
- I also gained significant experience with salesmanship, showing the cars to potential buyers as well as interacting and answering questions about the cars.

- Below are a few more regular/basic flips I took part in. Once I gained more hands-on experience with the cars as well as understanding how the process of registering and insuring vehicles works, I continued with selecting cars which required additional bodywork/repairs to get it in shape to sell. With these cars, I would get them at a cheaper price, but also be tasked with significantly harder issues which needed to be resolved.

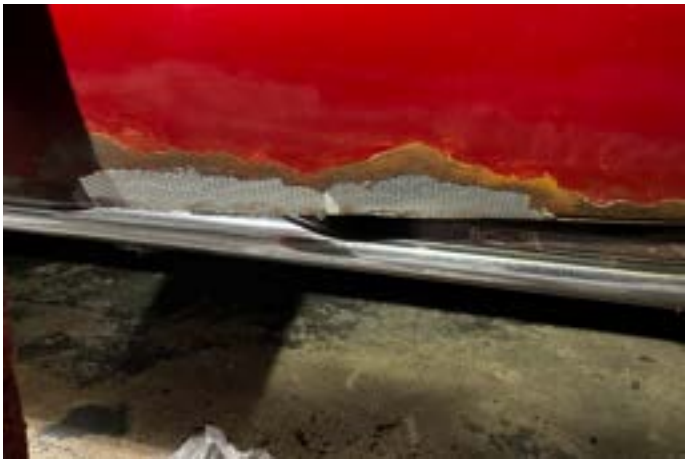


## Autobody/Bodywork

- The first car which involved a bit more work was this 2010 Dodge Ram 1500, which has significant rust/rot, as well as scuffing from the previous owner all along the side of the body as shown below.



- I used a rotary sander to wet sand the entire panel down to be prepped for the body filler. I used metal pieces to create a structure for the filler to mold around, sanded it down to be flush and finished with repainting and coating it with a clear coat.



- Overall, this experience helped me learn the basics with prepping a surface to repaint, this helped me later on with the next project which involved a bit more technical skill.

- This Lexus ES 300 was involved in a frontal collision, which was repairable, however it required extensive frame work, as well as the use of tools which granted the leverage to do so. The fender was removed, fixed, and suspended to be re-painted and fully restored to its original condition.

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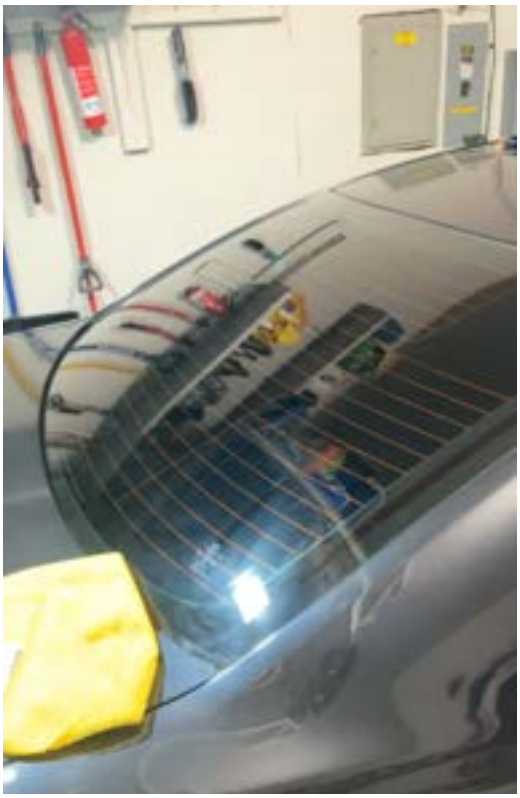
## Local Tinting/Wrapping and Maintenance/Bodywork

- After gaining the knowledge necessary to do the most common modifications and maintenance on cars, I expanded my reach to providing a local car service and modification business.
- I began by learning different modifications such as tinting, wrapping, and interior upholstery on my current car at the time (2013 Mazda 3). I did all of my offered modifications on my own car first to ensure that my quality of work was high enough to offer to clients.

- This is the first photo of the project car when I had first purchased it.

# Tinting

- First I began with basic taillight tinting, which many clients had requested. I had practiced on my own cars prior to accepting my first client. At the time, I was just getting familiar with vinyl so I opted for spray tint instead of vinyl.
- My first client was my uncle, who wanted just the red bit of the taillight smoked, which required a bit more masking and prep prior to spraying.
- Below is another client car which also had requested taillight tint



- Nearly every car which I had owned throughout the car flipping process, I had tinted all windows with 5% tint, as well as creating a sunstrip on the front windshield. I tinted my parents' cars, friends, and neighbours.

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## Upholstery/Interior Work

- My first interior upholstery involved taking the door panels off and wrapping the leather door inserts with alcantara-like suede material which required to be adhered and stitched to the door. Although an extensive task, the end result was nearly perfect.

- Next, I removed all of the seats from the interior to stitch made-to-fit seat covers to fully transform the interior of the car.

- This steering wheel was fully custom stitched, involving purchasing a roll of black perforated leather, measuring and cutting it to the correct diameter.

- Upon completing my personal vehicle, the owner of one of the cars which I had sold (2011 Lexus ES 300), had requested the same interior service, wanting the seats to be converted from a gray to a red interior.

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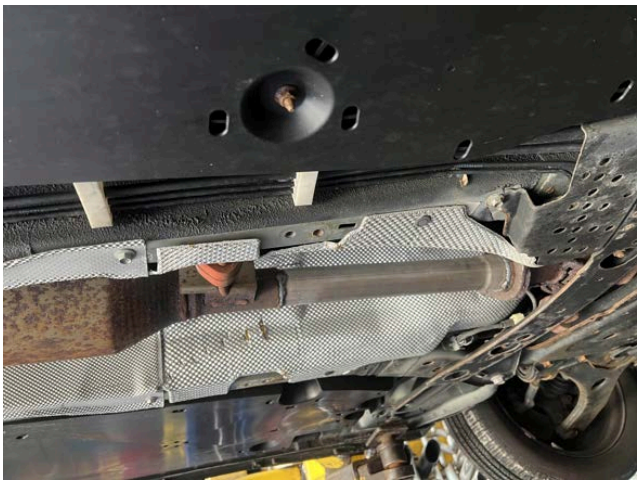
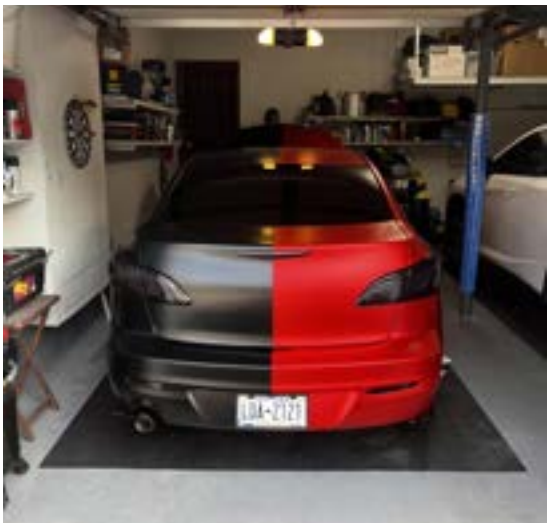
## Welding

- I have experimented with MIG welding at my cousin's autoshop, although I was being walked through the steps, I understood the basics of MIG welding.



## Wrapping

- Since I had experience with tinting, learning how to wrap was significantly easier as wrapping consists of the same basic principles as tinting.
- I chose to challenge myself and add an additional layer of complexity with my first wrap by doing a “half and half” color scheme, where the left side of the car would be a different color than the right, this involved using knifeless tape and advanced cutting techniques to ensure that the seam would be flush.



- After countless hours of trial and error, the car was complete. This was the final result.

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## Electrical Work

- This project involved wiring up a lightbar using an arduino to control strobe and brightness, it was installed flush within the front grill.

- This was a valved exhaust system which required the use of a programmed remote to control the butterfly valve, as well as hardwiring a switch and direct 12v power to the unit. This also had to be welded to the existing exhaust system.

- I have ample experience with wiring amps/speakers, integrated cameras, and aftermarket headunits, as I offered headunit installations to local clients.
- Below is the conversion of my personal vehicle.

# Arduino/3D Printing/Electronics

## Driving Simulator Project

- This driving simulator features a plethora of Arduino based electronics and add-ons, the most notable being the two fans mounted on top.
- The fans are controlled by an Arduino Uno running SIMHUB, an application which can convert in-game data from the simulator to CAN bus and other transmittable code. The fans are two PWM 140mm PC fans, and are directional. The loaded code can read the lateral and longitudinal g forces experienced within the game, and directly transmits that to the driver via the fans. In essence, if the car is accelerating in the rightward direction, then the left fan would spool up to simulate g forces. This significantly improves immersion when in VR, but can also be used standalone.

- This tachometer is wired up via an Arduino Nano, and reads in-game data and transmits it via CAN (Controller Area Network).
- Utilized a code conversion software to convert in-game rpm readings to transmittable code able to be read by the tachometer.